

A
Micro Project Report
On
Smart Industrial Air Quality & Leakage Guard

Submitted to
Department of Industrial IoT

In Partial Fulfillment of the degree of Bachelor of Technology

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CERTIFICATE

This is to certify that the Micro project entitled “**Smart Industrial Air Quality & Leakage**”

has been successfully presented by **Yug Ramteke ,Vaibhav Ghode ,Tushar Agashe ,Parth Janai** student of 4th semester IIoT for the partial fulfillment of the requirements for the Bachelor of Technology (B.Tech.) in Industrial IoT at St. Vincent Pallotti College of Engineering & Technology, Nagpur (Autonomous Institution).

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INDEX

Ch. No.	Title	Page No.
	Certificate	ii
	Acknowledgement	iii
	Abstract	v
	Table of Contents	vi
1	Introduction	1
	1.1 Background	1
	1.2 Objectives	2
	1.3 Scope of the Project	2
2	Literature Review	3
	2.1 Existing Systems	3
	2.2 Limitations of Existing Systems	3
3	Problem Statement	4
4	Methodology / System Design	5
	4.1 System Architecture	5
	4.2 Block Diagram	5
	4.3 Flowchart / Algorithm	6
5	Implementation	8
	5.1 Hardware Components	8
	5.2 Software Tools	11
	5.3 Circuit Diagram	11
6	Results and Discussion	12
	6.1 Output Screenshots	13
	6.2 Observations	14
7	Applications	15
8	Conclusion	16
9	Future Scope	17
10	References	18
11	Appendix	19

ABSTRACT

Industrial environments are highly prone to hazards caused by leakage of harmful and combustible gases such as LPG, methane, and smoke. These gases can lead to serious accidents including fire outbreaks, explosions, and health risks for workers. Therefore, there is a need for a smart system that can continuously monitor air quality and detect gas leakage at an early stage.

This project presents a Smart Industrial Air Quality and Leakage Guard system using Internet of Things (IoT) technology. The system is developed using an ESP32 microcontroller and an MQ-2 gas sensor. The sensor continuously monitors gas concentration in the surrounding environment and sends data to the ESP32 for processing.

When the gas level exceeds a predefined threshold, the system immediately activates a buzzer and displays a warning message on an LCD screen. This helps in providing early alerts to workers and prevents accidents. Additionally, the system can send data to a cloud platform for remote monitoring.

The proposed system is cost-effective, reliable, and easy to implement. It reduces human effort and improves industrial safety by providing real-time monitoring and instant alerts.

LIST OF FIGURES

Sr. No	Figure Name	Page No.
1	Block Diagram	5
2	Flowchart	7
3	Esp32	8
4	MQ2 and MQ135	8
5	LCD Display(16 x 2)	9
6	Buzzer	9
7	Breadboard	10
8	Jumper Wires	10
9	Circuit Diagram	11
10	Working Outputs	13

INTRODUCTION

In modern industrial environments, safety is one of the most important concerns due to the presence of hazardous gases and chemicals. Industries such as chemical plants, oil refineries, manufacturing units, and storage facilities frequently deal with combustible and toxic gases. Leakage of such gases can result in severe consequences, including fire hazards, explosions, environmental pollution, and health problems for workers.

Traditional gas detection systems are limited in functionality, as they mainly provide only local alerts such as alarms or indicators. These systems do not support continuous monitoring, remote access, or data storage, making them less effective in preventing accidents. In many cases, delayed detection or lack of proper warning systems leads to serious industrial disasters.

With the advancement of IoT technology, it has become possible to design smart systems that can monitor environmental conditions in real time. IoT-based systems can collect data from sensors, process it using microcontrollers, and transmit it to cloud platforms for remote monitoring. These features enable early detection, quick response, and better decision-making.

The **Smart Industrial Air Quality and Leakage Guard System** is developed to address these issues. It provides a reliable and automated solution for detecting gas leakage and monitoring air quality. By combining sensors, microcontrollers, and IoT technology, the system ensures safety and reduces the risk of accidents in industrial environments.

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1.2 Objectives

The main objectives of this project are:

1. To design a system that can continuously monitor air quality in industrial environments.
2. To detect leakage of harmful and combustible gases using gas sensors.
3. To provide immediate alerts through buzzer and display system when gas levels exceed safe limits.
4. To display real-time gas concentration data on an LCD screen.
5. To enable remote monitoring and data visualization using IoT platforms.
6. To reduce human effort in monitoring and improve system efficiency.
7. To enhance industrial safety and prevent accidents caused by gas leakage.

1.3 Scope of the Project

The scope of this project is focused on the detection and monitoring of gas leakage in industrial environments. The system can be deployed in areas where there is a risk of harmful gas release, such as factories, chemical plants, and storage facilities.

The system provides real-time monitoring and alert generation, ensuring quick response to hazardous conditions. It is designed to be cost-effective and scalable, allowing it to be implemented in multiple locations.

However, the project is limited to detection and alert mechanisms only. It does not include advanced control features such as automatic shutdown of gas supply or ventilation systems. These features can be added in future enhancements to improve the system's functionality.

2. LITERATURE REVIEW

2.1 Existing Systems

In industrial environments, safety systems for detecting gas leakage have been in use for many years. Most traditional systems rely on basic gas sensors connected to alarm units. These systems are designed to detect the presence of harmful gases and trigger an audible alert such as a buzzer or siren when the gas concentration exceeds a certain limit.

In some industries, manual monitoring methods are still used, where workers periodically check gas levels using handheld detectors. Additionally, certain advanced setups use fixed gas detection systems integrated with control panels that can activate ventilation systems or shut down operations during emergencies.

With the advancement of technology, some modern systems have started incorporating microcontrollers and wireless communication. These systems allow limited automation and can display sensor readings on local screens. However, many of these systems still lack real-time remote monitoring and cloud integration.

Overall, existing systems focus mainly on detection and local alerting, but they do not provide a complete smart solution for continuous monitoring, data storage, and remote accessibility.

2.2 Limitations of Existing Systems

Despite their usefulness, existing gas detection systems have several limitations:

One of the major drawbacks is the lack of real-time monitoring. Traditional systems do not continuously send data to a centralized platform, making it difficult to track changes over time.

Another limitation is the absence of remote access. Most systems provide only local alerts, which means that authorities or supervisors cannot monitor the system from a remote location.

Many systems also depend on manual intervention, which increases the chances of human error. Delays in response time can lead to serious accidents.

Additionally, these systems often do not store historical data, making it difficult to analyze trends or predict future risks.

Power dependency is another issue. During power failures, many systems stop functioning, reducing reliability in critical situations.

Lastly, existing systems are not easily scalable and are not designed for integration with smart industrial infrastructure.

3. PROBLEM STATEMENT

Industrial areas are highly prone to hazards caused by gas leakage and poor air quality. Harmful gases such as LPG, methane, carbon monoxide, and smoke can accumulate in industrial environments due to leakage, incomplete combustion, or equipment failure.

The presence of these gases poses serious risks, including fire hazards, explosions, health problems, and environmental damage. In many cases, gas leakage goes undetected until it reaches dangerous levels, leading to accidents and financial losses.

Traditional detection systems are not efficient enough to handle these challenges, as they lack real-time monitoring, remote accessibility, and automated alert mechanisms. Workers often depend on manual monitoring or delayed alerts, which increases the risk of accidents.

Therefore, there is a need for a smart, automated, and reliable system that can continuously monitor air quality, detect gas leakage at an early stage, and provide immediate alerts to ensure safety and prevent industrial hazards.

Traditional safety methods often rely on manual inspections or localized, non-smart detectors that fail to provide real-time data or remote alerts. Invisible Threats: Many industrial gases (like LPG, CO, and NH₃) are colourless, making them impossible for workers to detect until they reach toxic or explosive concentrations. Delayed Response: Standard alarm systems often lack automated logic to provide specific feedback (like "Poor Air Quality" vs. "Gas Leakage"), leading to confusion during emergencies. Lack of Continuous Monitoring: Without an automated system, intermittent leakages can go unnoticed for long periods, causing chronic respiratory issues for workers and increasing the risk of fire hazards. Infrastructure Gaps: Most small-to-medium-scale industries lack an integrated, low-cost "guard" system that can simultaneously monitor air quality and gas leaks while providing clear visual feedback on an LCD.

4. METHODOLOGY / SYSTEM DESIGN

4.1 System Architecture

The Smart Industrial Air Quality and Leakage Guard system is designed using a simple yet effective architecture consisting of input, processing, and output units.

The input unit includes the MQ-2 gas sensor, which continuously monitors the concentration of gases in the surrounding environment. The sensor detects gases such as LPG, methane, hydrogen, and smoke.

The processing unit consists of the ESP32 microcontroller, which receives data from the sensor. The ESP32 processes this data, compares it with predefined threshold values, and decides whether the environment is safe or hazardous.

The output unit includes a buzzer and an LCD display. The LCD shows real-time gas levels, while the buzzer provides an alert when gas concentration exceeds safe limits.

Additionally, the system can be connected to an IoT platform for remote monitoring and data visualization.

4.2 Block Diagram

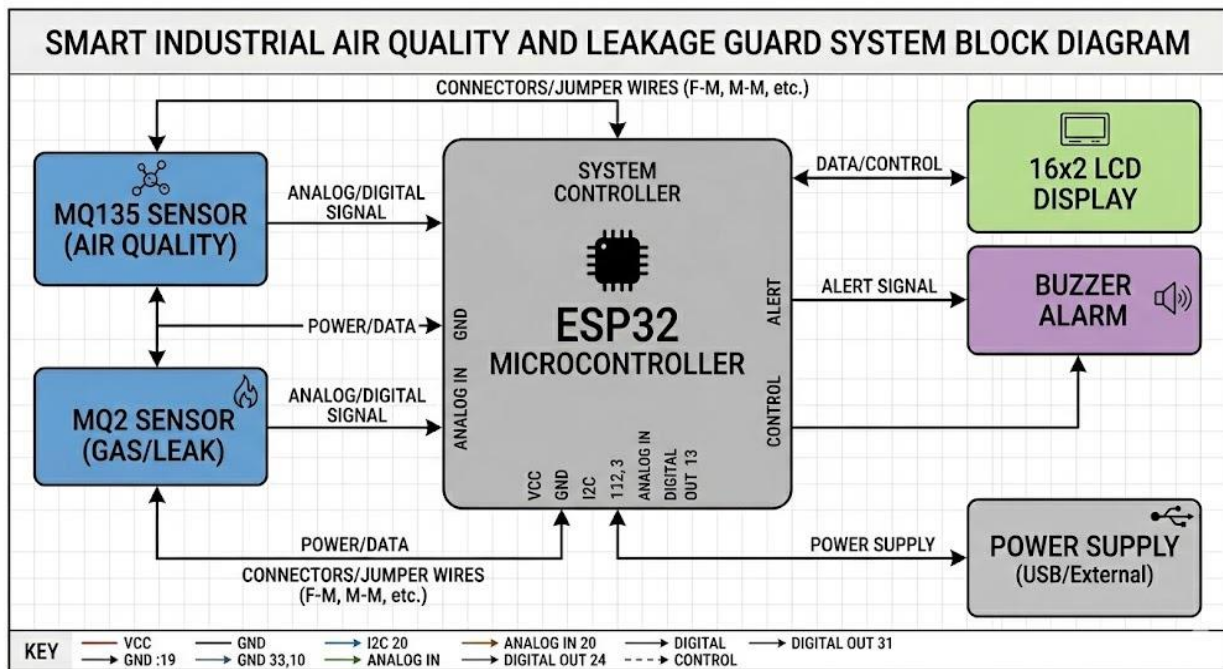


Fig. No 1 : Block Diagram

4.3 Flowchart / Algorithm

The system follows a step-by-step process:

Step 1: Start

Step 2: Initialize the ESP32 system

Step 3: Initialize sensors

MQ135 Gas Sensor

MQ2 Gas Sensor

Step 4: Initialize 16x2 LCD Display

Step 5: Set threshold values

Air quality limit (MQ135)

Gas leakage limit (MQ2)

Step 6: Begin loop

Step 7: Read air quality data from MQ135

Step 8: Read gas leakage data from MQ2

Step 9: Display both values on LCD

Step 10: Compare MQ135 value with threshold

If value > limit → Display “Poor Air Quality”

Else → Display “Air Quality Normal”

Step 11: Compare MQ2 value with threshold

If value > limit → Display “Gas Leakage Detected”

Else → Display “No Leakage”

Step 12: If any condition is unsafe

Show warning message on LCD

(Optional) Send alert using ESP32 Wi-Fi

(Optional) Activate buzzer

Step 13: Wait for short delay

Step 14: Repeat from Step 7 (continuous monitoring)

Step 15: End

Flowchat:

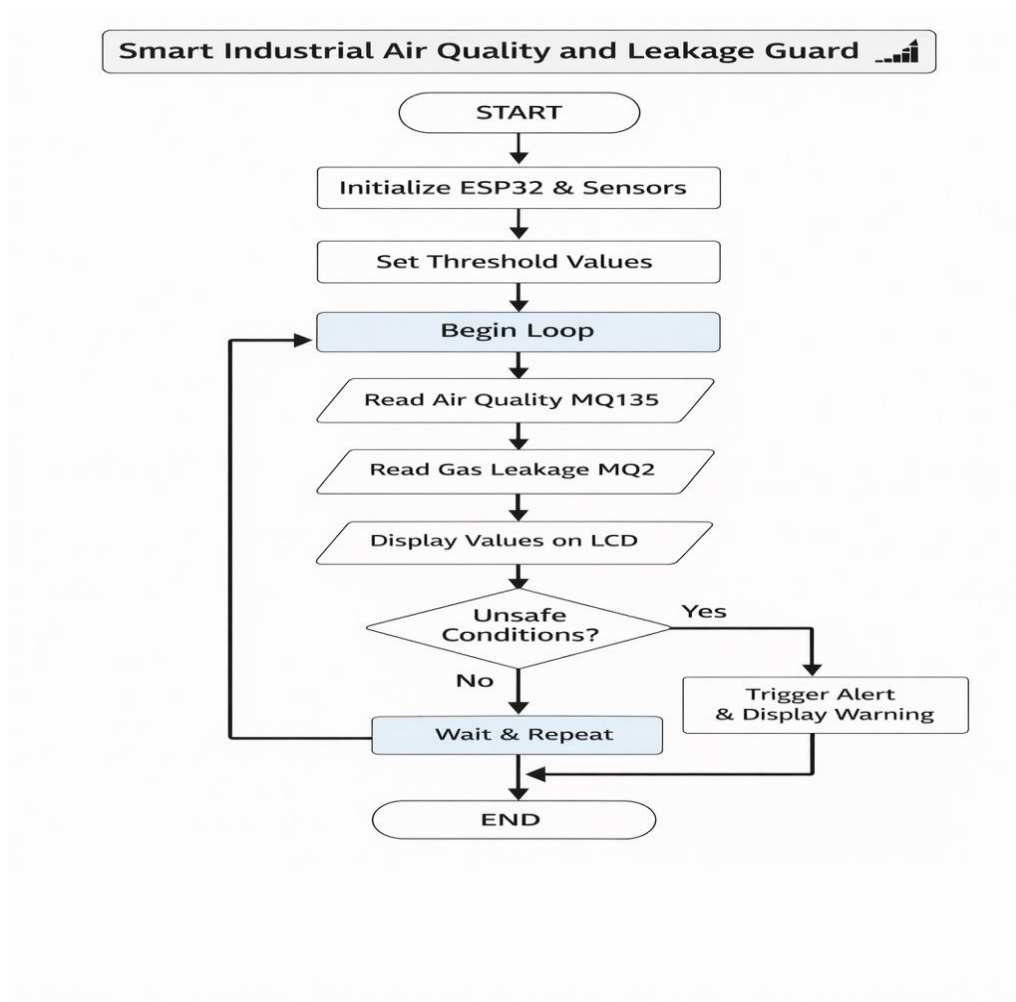


Fig. No 2 : Flowchart

5. IMPLEMENTATION

5.1 Hardware Components

1. ESP32 Microcontroller

ESP32 is the main controller used in the system. It processes sensor data and controls output devices. It also supports Wi-Fi, making it suitable for IoT applications.

Fig. No 3 : Esp32



2. MQ-2 and MQ-135 Gas Sensor

This sensor detects gases like LPG, methane, hydrogen, and smoke. It provides analog output based on gas concentration.

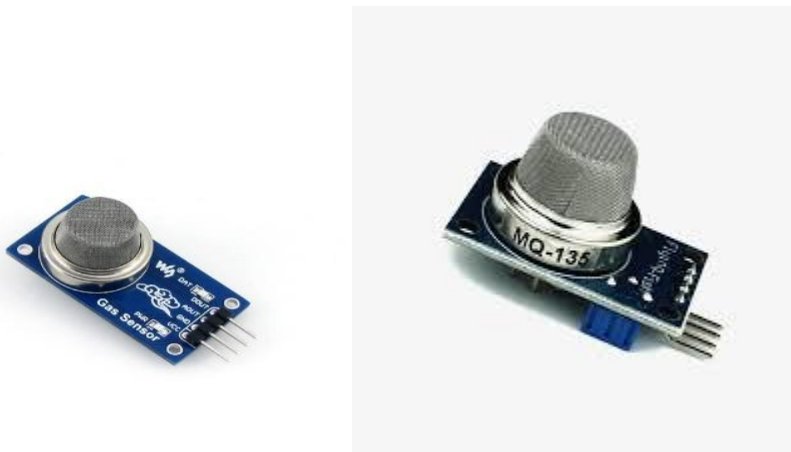


Fig. No 4 : MQ2 and MQ135

3. LCD Display (16x2)

Used to display real-time gas levels and system status.



Fig. No 5 : LCD Display(16 x 2)

4. Buzzer

Acts as an alert system to warn users when gas leakage is detected.



Fig. No 6 : Buzzer

5. Breadboard

Used for circuit connections and testing before PCB design.

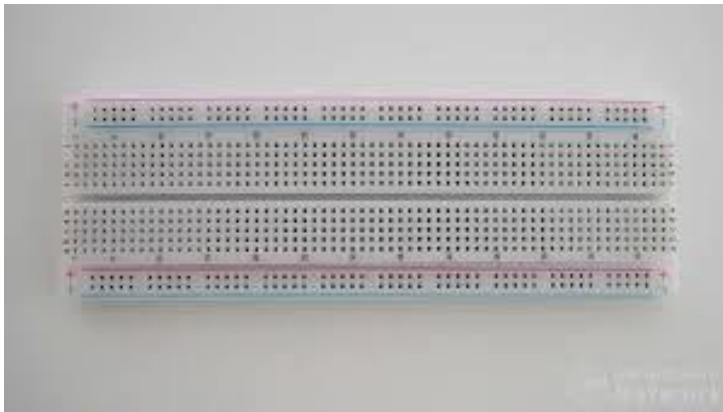


Fig. No 7 : Breadboard

6. Jumper Wires

Jumper wires are used to establish electrical connections between different components such as the ESP32, sensor, and relay module. They are essential for circuit setup and prototyping.



Fig. No 8 : Jumper Wires

5.2 Software Tools

- **Arduino IDE** : Arduino IDE is used as the primary development environment for writing, compiling, and uploading code to the ESP32 microcontrollers. It provides a user-friendly interface and supports programming in C/C++. In this project, it is used to implement the logic for ultrasonic sensing, water level calculation, threshold comparison, and ESP-NOW communication between nodes.
- **Embedded C/C++** : The system logic is implemented using Embedded C/C++ programming language within the Arduino IDE. It includes sensor interfacing, data processing, wireless communication setup, and control of output devices like the relay module.

5.3 Circuit Diagram

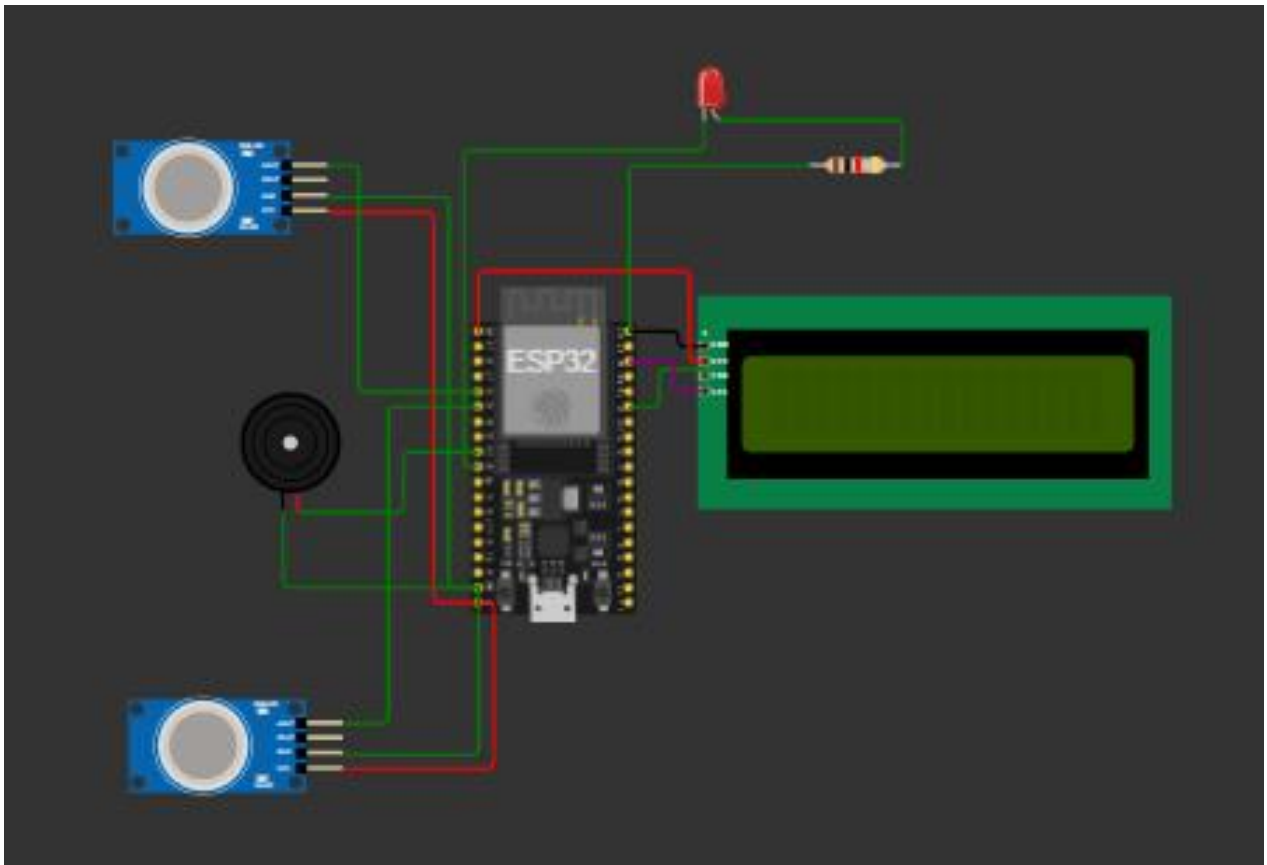


Fig. No 9 : Circuit Diagram

6. RESULTS AND DISCUSSION

The Smart Industrial Air Quality and Leakage Guard system was successfully implemented and tested using a breadboard setup before PCB design. The system was evaluated under different environmental conditions to analyze its performance in detecting gas leakage and monitoring air quality.

During the testing phase, the MQ-2 gas sensor was exposed to different gas concentrations such as smoke and LPG to simulate leakage conditions. The sensor continuously monitored the air and transmitted real-time data to the ESP32 microcontroller. The ESP32 processed the data and compared it with predefined threshold values to determine whether the environment was safe or hazardous.

The system demonstrated reliable performance in detecting gas leakage. When the gas concentration remained within the safe limit, the system operated normally and displayed the gas levels on the LCD screen. However, when the gas concentration exceeded the threshold value, the system immediately responded by activating the buzzer and displaying a warning message on the LCD.

The integration of the LCD display provided a clear and real-time indication of the gas levels, while the buzzer ensured that immediate attention was drawn to dangerous situations. The system response time was observed to be fast, which is crucial in preventing accidents in industrial environments.

Overall, the system performed efficiently and met the expected objectives of real-time monitoring, early detection, and alert generation.

6.1 Output Screenshots

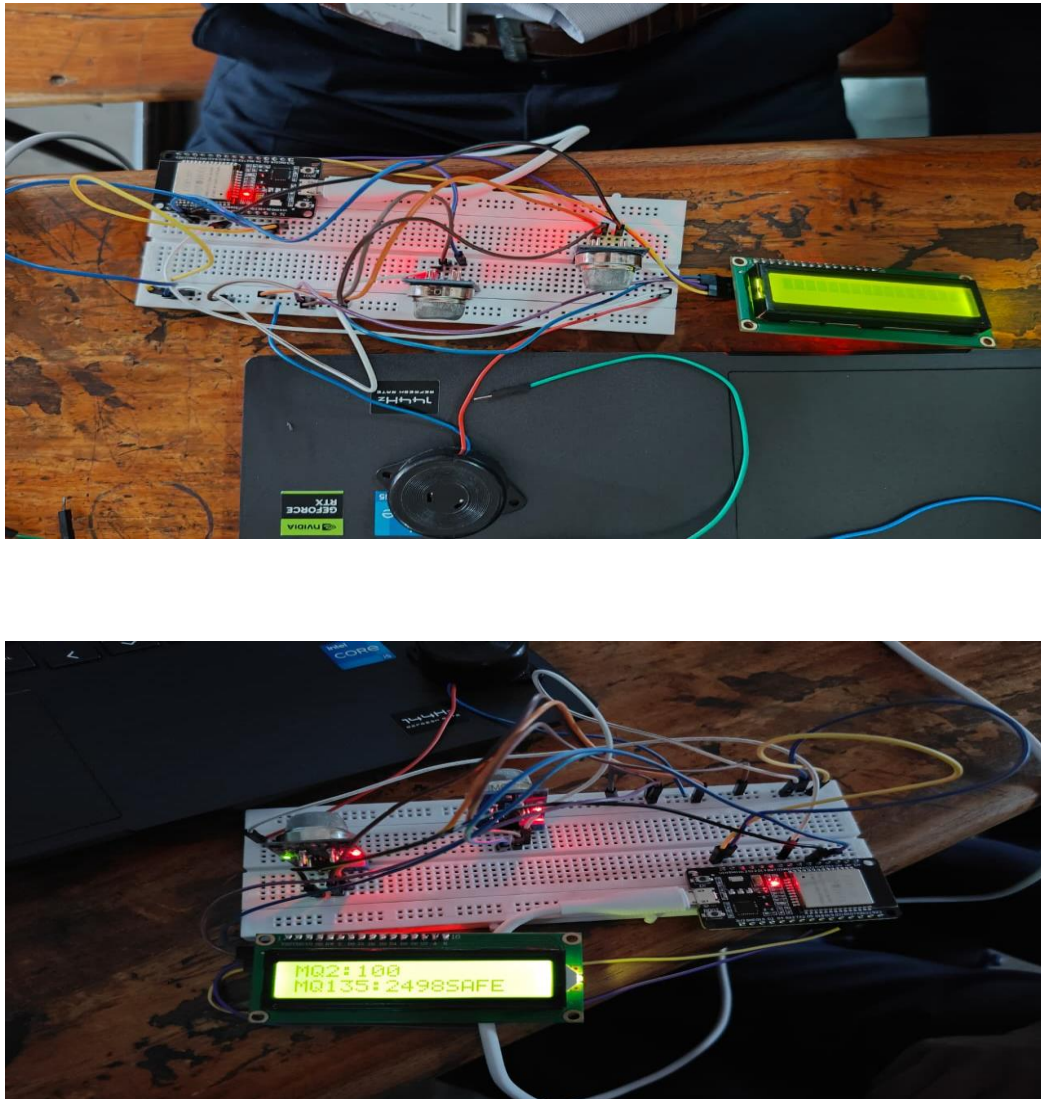


Fig. No 10 : Working Outputs

The above figure shows the output of the Smart Industrial Air Quality and Leakage Guard system during testing. The setup consists of an ESP32 microcontroller connected to an MQ-2 gas sensor, LCD display, and buzzer. The LCD screen displays real-time gas concentration levels, while the buzzer provides an alert when the gas concentration exceeds the safe threshold.

During normal conditions, the LCD shows safe gas levels, indicating that the environment is secure. When gas leakage is detected, the system immediately activates the buzzer and displays a warning message, alerting users about the hazardous condition.

6.2 Observations

During the implementation and testing of the system, several important observations were made:

The MQ-2 gas sensor showed a **quick response** to the presence of gases such as smoke and LPG. The sensor was able to detect even small changes in gas concentration, making it suitable for real-time monitoring applications.

The ESP32 microcontroller processed the sensor data efficiently and triggered the output devices without noticeable delay. This ensured that the system provided **instant alerts**, which is essential for safety-critical applications.

The LCD display worked effectively in showing real-time gas levels and system status, making it easy for users to monitor the condition of the environment. The buzzer provided a **clear and immediate warning signal**, ensuring that users were alerted even without looking at the display.

The system was found to be **reliable and stable** during continuous operation. However, it was observed that the sensor requires proper calibration for accurate readings, and the system depends on a continuous power supply for uninterrupted functioning.

Overall, the system successfully met its objectives and proved to be an effective solution for gas leakage detection and air quality monitoring.

7. APPLICATIONS

The Smart Industrial Air Quality and Leakage Guard system can be used in various areas where gas leakage and air pollution are major concerns. It is mainly useful in **chemical industries** and **oil and gas plants**, where harmful and combustible gases are handled regularly.

The system can also be applied in **manufacturing units** to maintain a safe working environment by continuously monitoring air quality. In **storage tanks and pipelines**, it helps in detecting gas leakage at an early stage, preventing damage and accidents.

Additionally, it can be used in **laboratories** where gases are used during experiments, and even in **residential areas** for detecting LPG leakage in kitchens. Overall, the system helps in improving safety and reducing risks in both industrial and domestic environments.

Industrial Safety Monitoring

Used in factories and industries to detect harmful gases and prevent accidents due to gas leakage.
Mining Industry

Detects dangerous gases in mines, preventing suffocation and explosions.
Smart Buildings

Used in offices, malls, and buildings to monitor indoor air quality and gas leakage.
Environmental Monitoring

Helps track pollution levels and maintain environmental standards.

Laboratories

Detects gas leaks and maintains safe conditions in research labs.

Home Safety Systems (Extension)

Can be used in homes for LPG gas leakage detection and air quality monitoring.

8. CONCLUSION

The Smart Industrial Air Quality and Leakage Guard system is a reliable solution for detecting gas leakage and monitoring air quality in real time. The system uses an MQ-2 gas sensor and ESP32 microcontroller to continuously monitor gas levels and provide alerts when unsafe conditions are detected.

When the gas concentration exceeds the safe limit, the system activates a buzzer and displays a warning on the LCD, ensuring quick response to hazardous situations. The project is simple, cost-effective, and easy to implement.

Overall, this system helps in improving industrial safety, reducing human effort, and preventing accidents caused by gas leakage.

"In conclusion, the project successfully met its objectives of real-time air quality assessment and gas leakage detection. The system demonstrated high sensitivity in detecting pollutants using the MQ135 and combustible gases via the MQ2. The logic implemented ensures that once pre-set safety thresholds are crossed, the system triggers immediate defensive actions. The use of the ESP32 provides a robust platform for data processing, confirming that affordable sensor technology can be reliably deployed for critical safety guarding in an industrial environment." This project concludes that smart, sensor-driven safety guards are essential for modern industrial operations. By leveraging the ESP32's processing power and sensitive gas detection modules, we have created a reliable 'guard' that minimizes human error in leak detection. This system serves as a foundational model for Industry 4.0 safety protocols, where automated monitoring and instant alerting become the standard for protecting both personnel and infrastructure."

9. FUTURE SCOPE

The system can be further improved by adding advanced features to enhance its performance. An **automatic gas shutoff system** can be added to stop leakage instantly. A **mobile application** can be developed to send real-time notifications to users.

The system can also be upgraded by using **multiple sensors** to detect different types of gases. Integration with **IoT platforms** for better data analysis and monitoring can make the system more efficient. In the future, technologies like **AI and machine learning** can be used to predict gas leakage and prevent accidents in advance.

Predictive Maintenance: Instead of just reacting to a leak, you could use ML algorithms to analyze historical sensor data. This helps predict when a pipe might fail or when a sensor needs calibration before a dangerous situation occurs.

False Alarm Reduction: AI can help the system distinguish between a real gas leak and a momentary spike caused by cleaning agents or exhaust from a passing vehicle, reducing unnecessary factory shutdowns. **Local Processing:** By moving more "intelligence" to the ESP32 (the "Edge") rather than relying entirely on the Cloud, the system can make split-second decisions (like shutting off a valve) even if the Wi-Fi goes down.

Latency Reduction: In industrial safety, every millisecond counts. Edge computing ensures the alarm sounds the moment the sensor detects a ppm spike. **Mobile Guarding:** For large industrial plants, fixed sensors have "blind spots." Future versions of your project could involve mounting these sensor modules on autonomous drones that patrol the facility to find leaks in high-up pipes or remote areas.

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APPENDIX

❖ PROGRAM CODE (ESP32)

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// □ WiFi
const char* ssid = "OnePlus Nord 4";
const char* password = "123451234";
// □ Telegram
String botToken = "8747945819:AAH_VZwVTt5di1EB_IZ8OxZGqueZiE9G6WQ";
String chatID = "607993327";
// □ LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);
// □ Pins
#define MQ2_PIN 34
#define MQ135_PIN 35
#define BUZZER_PIN 25
#define LED_PIN 26
// □ Separate thresholds
int mq2Threshold = 100; // adjust after testing
int mq135Threshold = 50000; // adjust after testing
unsigned long lastSend = 0;
unsigned long interval = 10000; // 10 sec
// □ Telegram function
void sendTelegram(String msg) {
  if (WiFi.status() == WL_CONNECTED) {
    HTTPClient http;
    String url = "https://api.telegram.org/bot" + botToken +
      "/sendMessage?chat_id=" + chatID +
      "&text=" + msg;
```

```

    http.begin(url);
    http.GET();
    http.end();
  }
}

void setup() {
  Serial.begin(115200);
  lcd.init();
  lcd.backlight();
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
  // WiFi connect
  lcd.print("Connecting WiFi");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  lcd.clear();
  lcd.print("WiFi Connected");
  delay(2000);
  lcd.clear();
  sendTelegram("System Started ☑");
}

void loop() {
  int mq2 = analogRead(MQ2_PIN);
  int mq135 = analogRead(MQ135_PIN);
  Serial.print("MQ2: ");
  Serial.print(mq2);
  Serial.print(" | MQ135: ");
  Serial.println(mq135);
  // LCD display

```

```

lcd.setCursor(0,0);
lcd.print("MQ2:");
lcd.print(mq2);
lcd.print(" ");
lcd.setCursor(0,1);
lcd.print("MQ135:");
lcd.print(mq135);
lcd.print(" ");
String status = "SAFE ✔";
// □ Individual detection
if (mq2 > mq2Threshold) {
    status = "MQ2 ALERT □";
}
if (mq135 > mq135Threshold) {
    status = "MQ135 ALERT □";
}
// Combined alert
if (mq2 > mq2Threshold || mq135 > mq135Threshold) {
    lcd.setCursor(10,1);
    lcd.print("ALERT");
    digitalWrite(BUZZER_PIN, HIGH);
    digitalWrite(LED_PIN, HIGH);
} else {
    lcd.setCursor(10,1);
    lcd.print("SAFE ");

    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_PIN, LOW);
}
// □ Send Telegram update
if (millis() - lastSend > interval) {
    String msg = "Gas Update:\n";

```

```
msg += "MQ2: " + String(mq2) + "\n";  
msg += "MQ135: " + String(mq135) + "\n";  
msg += "Status: " + status;  
sendTelegram(msg);  
lastSend = millis();  
}  
delay(1000);  
}
```